Notes: Li, Mai, Shen, Yang, and Zhang (RFS, 2026)

Dissecting Corporate Culture Using Generative AI

Contents

1 Big picture

- **Question.** How do different corporate stakeholder groups—analysts, managers, and employees—understand corporate culture, and how do these differences affect financial decisions and market prices?
- **Main innovation.** The paper applies generative AI as a reasoning agent to analyst reports, earnings call transcripts, and Glassdoor employee reviews. The AI extracts structured cause–effect relations involving corporate culture.
- **Main data scale.** The paper analyzes 2.4 million analyst reports from 2000–2020, 243 thousand earnings calls from 2004–2020, and 5.3 million Glassdoor reviews from 2008–2020.
- **Main methodological output.** The authors build a knowledge graph linking culture types to their perceived causes and business effects. Each relation is canonicalized into standardized categories and assigned a tone.
- **Core stakeholder finding.** Analysts, managers, and employees emphasize different cultural dimensions because they occupy different roles and face different incentives.
- **Analysts' view.** Analysts emphasize innovation and adaptability, collaboration and people-focused culture, and especially integrity and risk management more than internal stakeholders do.
- **Managers' view.** Management discussions emphasize strategic causes, such as business strategy, customer relations, and disruptive technologies, and link culture to market share, growth, and profitability.
- **Employees' view.** Employee reviews emphasize collaboration and people-focused culture, management teams, employee hiring and retention, internal conflicts, and employee satisfaction.
- **Validation finding.** AI-extracted culture tones correlate with external markers of the corresponding culture type: workforce scores, consumer complaints, brand value, patents, R&D, restatements, risk, ROA, and sales growth.
- **M&A application.** Analysts' culture-related discussion of M&A deals predicts acquirer announcement returns, and negative market reactions make analysts more likely to discuss culture and M&As after deal announcements.
- **Analyst-output finding.** Analysts' tones about culture are incorporated into stock recommendations and target prices, especially when reports contain deeper cause–effect analysis.
- **Market-reaction finding.** Investors react to analysts' culture tone. The reaction is stronger when the report contains many cause/effect relations or when the analyst's view differs from management and employee views.
- **Economic takeaway.** Generative AI can convert qualitative corporate culture narratives into structured, economically meaningful measures. Analysts' differentiated cultural assessments are not mere text sentiment: they inform recommendations, target prices, and investor reactions.

2 Data and measurement

2.1 Text corpora and sample construction

- **Analyst reports.** The main sample contains more than 2.4 million analyst reports from Thomson One's Investext database, covering S&P 1500 firms over 2000–2020.
- **Earnings calls.** The management corpus contains 243,501 earnings conference calls from Capital IQ Transcripts over 2004–2020.
- **Employee reviews.** The employee corpus contains 5.3 million Glassdoor reviews from Revelio Labs over 2008–2020.

- **Report metadata.** For analyst reports, the authors obtain report date, gvkey, lead analyst name, and broker name.
- **Text conversion.** Reports are converted from PDF to plain text. Because conversion can destroy paragraph structure, the paper uses the C99 text segmentation algorithm to reconstruct coherent text segments.
- **Boilerplate filter.** A BERT classifier is trained to identify boilerplate report segments. The authors retain only segments with predicted boilerplate probability at or below the sample median.

Interpretation: The paper is not simply running sentiment analysis on all text. It first reconstructs meaningful segments and removes boilerplate so that the AI model reasons about economically relevant content.

2.2 Finding culture-related segments

The authors use a three-stage procedure to identify segments that substantively discuss corporate culture:

1. **Keyword search.** Find text containing explicit culture-related terms such as "corporate culture" and "workplace culture."
2. **BERT semantic classifier.** Identify segments that discuss culture without using explicit culture keywords.
3. **Generative AI filter.** Use GPT to keep only candidate segments that substantively analyze organizational culture.

The final analyst-report data set contains 138,545 culture-related segments in 86,112 reports.

Interpretation: The hybrid extraction method matters because analysts often discuss culture indirectly through practices, values, employee relations, governance, strategy, and firm behavior without using the word "culture."

2.3 Culture types, causes, effects, and tones

The paper asks the generative AI model to extract and standardize several objects:

- **Culture type:** one of six categories: collaboration and people-focused, customer-oriented, innovation and adaptability, integrity and risk management, performance-oriented, and miscellaneous.
- **Cause:** one of 18 standardized factors that may shape culture, such as business strategy, management teams, employee hiring and retention, M&As, regulatory issues, or customer relations.
- **Effect:** one of 17 standardized outcomes affected by culture, such as market share and growth, profitability, customer satisfaction, risk management, innovation, employee satisfaction, misconduct, and internal conflict.
- **Tone:** whether the culture-related discussion is positive, neutral, or negative.
- **Triple:** a structured cause–effect relation linking a culture type to a cause or effect, with direction and tone.

Interpretation: The key measurement contribution is that the paper extracts reasoning, not just word counts. A statement is turned into a causal triple such as "focused selling culture results in double digit earnings growth."

2.4 Canonicalization and knowledge graph

After extracting raw phrases, the paper standardizes them:

- Extracted causes and effects are embedded into a vector space.
- The authors group raw phrases into 50 clusters.
- Manual review maps clusters into 18 causes and 17 effects.
- GPT then maps each raw extracted phrase to the closest standardized entity.
- Each standardized relation enters a knowledge graph linking causes, culture types, and effects.

Interpretation: Canonicalization is necessary because analyst language varies widely. Without standardization, the knowledge graph would be too sparse to compare stakeholders or conduct regressions.

2.5 Chain-of-thought prompting and retrieval augmented generation

The extraction prompt uses two important additions:

- **Chain-of-thought prompting:** the model answers a sequence of questions about culture type, tone, causes, effects, and causal relations before outputting a final structured JSON triple.
- **Retrieval augmented generation (RAG):** if the model says it needs more context, the authors retrieve other culture-relevant segments from the same report and provide them as additional input.

Interpretation: CoT improves the model's reasoning about causal relations, and RAG helps when a single segment lacks enough context. This design is important because cause–effect extraction is more difficult than classifying sentiment.

2.6 External validation variables

The paper validates the extracted culture measures using external markers:

- Collaboration and people-focused culture: workforce scores and employee overall ratings.
- Customer-oriented culture: consumer complaints and brand value.
- Innovation and adaptability culture: patents and R&D spending.
- Integrity and risk management culture: restatements and operating performance volatility.
- Performance-oriented culture: ROA and sales growth.

Interpretation: These tests ask whether the AI-extracted culture tones align with real-world markers of the corresponding organizational practices and outcomes.

3 Models and empirical specifications

3.1 Generative AI extraction algorithm

Conceptually, the extraction process can be written as

$$TextSegment_{i,t} \xrightarrow{GPT} \{CultureType, Cause, Effect, Tone, Direction\}_{i,t}. \tag{1}$$

At the report level, extracted triples are aggregated into firm-year, analyst-report, and stakeholder-corpus measures:

$$CultureMeasure^{g}_{i,t} = Aggregate\left(Triples^{g}_{i,t}\right), \tag{2}$$

where g indexes stakeholder group: analyst reports, earnings calls, or employee reviews.

Interpretation: The paper treats generative AI as a structured information-extraction engine. The resulting variables can be used in standard finance regressions.

3.2 Validation regressions

The validation specification is

$$Outcome_{i,t} = \alpha + \beta\, CultureTone_{i,t-1} + \gamma' Controls_{i,t-1} + IndustryFE + YearFE + \varepsilon_{i,t}. \tag{3}$$

The outcome depends on which culture type is being validated: employee outcomes for collaboration, brand/complaints for customer orientation, patents/R&D for innovation, restatements/risk for integrity, and performance for performance-oriented culture.

Interpretation: A positive or correctly signed β supports the claim that the extracted tone captures a meaningful corporate culture dimension.

3.3 M&A price-reaction regressions

The M&A testing regression is

$$CAR[-1,+1]_d = \alpha + \beta\, PrebidCultureTone_d + \delta' Controls_d + IndustryFE + YearFE + \varepsilon_d, \tag{4}$$

where d indexes deals.

The paper also separates M&A as a cause of cultural change from M&A as an effect of culture:

$$CAR[-1,+1]_d = \alpha + \beta_1 Tone_d^{M\&A\ as\ cause} + \beta_2 Tone_d^{M\&A\ as\ effect} + \delta' Controls_d + \varepsilon_d. \tag{5}$$

Interpretation: These regressions test whether analysts' culture analyses around deals contain information about market reactions to acquisitions.

3.4 Determinants of analyst culture coverage

At the firm-year level, the paper estimates

$$Y_{i,t} = \alpha + \beta' FirmCharacteristics_{i,t-1} + Ind/FirmFE + YearFE + \varepsilon_{i,t}, \tag{6}$$

where $Y_{i,t}$ is whether analysts discuss culture or whether analysts discuss a specific culture type.

At the firm-analyst-year level, the paper estimates

$$Y_{i,j,t} = \alpha + \beta' AnalystCharacteristics_{j,t-1} + Firm \times YearFE + BrokerFE + \varepsilon_{i,j,t}, \tag{7}$$

where $Y_{i,j,t}$ captures whether an analyst discusses culture, the number of culture types, the number of causes/effects, or tone.

Interpretation: These regressions ask when analysts choose to analyze culture and which analysts are more likely to perform deeper culture analysis.

3.5 Divergence across stakeholder perspectives

The paper constructs divergence measures between analysts and internal stakeholders. In general form,

$$Divergence_{i,t}^{A,M} = Distance\left(CultureProfile_{i,t}^{Analyst}, CultureProfile_{i,t}^{Management}\right), \tag{8}$$

$$Divergence_{i,t}^{A,E} = Distance\left(CultureProfile_{i,t}^{Analyst}, CultureProfile_{i,t}^{Employee}\right). \tag{9}$$

The profiles can be based on discussion frequencies over culture types, causes, effects, or tones.

Interpretation: Divergence measures quantify how far analysts' external assessments are from management's or employees' internal narratives.

3.6 Analyst recommendations and target prices

At the report level, the paper estimates specifications of the form

$$Output_r = \alpha + \beta_1 Tone_r + \beta_2 (Tone_r \times Depth_r) + \beta_3 Depth_r + \gamma' Controls_r + FE + \varepsilon_r, \tag{10}$$

where $Output_r$ is a recommendation upgrade or target price revision and $Depth_r$ is the number of culture-related causes/effects extracted from the report.

Interpretation: The coefficient on $Tone$ tests whether analysts incorporate culture tone into their recommendations and target prices. The interaction tests whether deeper causal analysis makes culture tone more decision-relevant.

3.7 Market-reaction regressions

The market-reaction regression is

$$CAR[-1,+1]_r = \alpha + \beta Tone_r + \gamma' Controls_r + FE + \varepsilon_r. \tag{11}$$

The heterogeneous effect specification is

$$CAR[-1,+1]_r = \alpha + \beta_1 Tone_r + \beta_2 (Tone_r \times Divergence_r) + \beta_3 Divergence_r + \gamma' Controls_r + FE + \varepsilon_r. \tag{12}$$

Interpretation: These regressions test whether investors react to analysts' culture narratives, especially when analysts provide independent assessments that diverge from internal stakeholder views.

4 Identification and empirical design

4.1 Validation logic

- If GPT-extracted culture tones are meaningful, they should align with independent, external measures of the corresponding culture type.
- The validation tests use outcomes that are not the same text as the analyst reports.
- Correctly signed associations reduce concern that the AI is merely producing arbitrary labels.

Interpretation: Validation is essential because the paper uses generated variables. The external marker tests show that extracted culture categories map to real corporate practices and outcomes.

4.2 Stakeholder comparison logic

- Analyst reports, earnings calls, and employee reviews are very different corpora.
- Applying a common taxonomy to all three corpora makes stakeholder comparisons possible.
- Differences in emphasis are interpreted through stakeholder roles: analysts monitor and value firms, managers justify strategy, and employees report lived workplace experience.

Interpretation: The stakeholder comparison is descriptive but powerful because it shows systematic role-consistent differences across three independent text sources.

4.3 Price formation logic

- Analyst reports have known release dates.
- Report-level culture tone can be linked to recommendation revisions, target price revisions, and stock returns around report release.
- Controls include nonculture report tone, report length, earnings forecast revisions, recommendation revisions, target price revisions, prior CAR, other analyst/firm controls, and fixed effects.

Interpretation: The paper separates culture-specific tone from general report sentiment by controlling for tone in nonculture sections of the report.

4.4 What the paper can and cannot prove

- The paper can show that extracted culture measures align with external culture markers and are incorporated into analyst output and price reactions.
- It cannot prove that culture itself causally changes firm outcomes in every setting.
- It cannot prove that generative AI perfectly understands all analyst text; the model performance tests show high but imperfect accuracy.
- It cannot fully separate analysts' independent information from information revealed by firms, although divergence tests help.

Interpretation: The strongest claim is that generative AI can extract structured, economically meaningful culture narratives and that those narratives matter for analyst decisions and investor reactions.

5 Tables and figures

5.1 Figure 1 and Table 1: extraction method and prompts

Read:

- Figure 1 summarizes the pipeline: start from 2.4 million analyst reports, reconstruct text segments, filter boilerplate, identify culture segments, apply GPT, extract triples, and validate outputs.
- Table 1 contains the main prompt and canonicalization prompt used for culture type, causes, effects, tone, and JSON triple extraction.
- The prompt forces the model to identify a culture type, assess tone, determine whether causal analysis exists, and then extract structured relationships.
- RAG is used only when the model requests more context.

Economic message: The methodology is built to extract reasoning about culture, not just sentiment or word frequency.

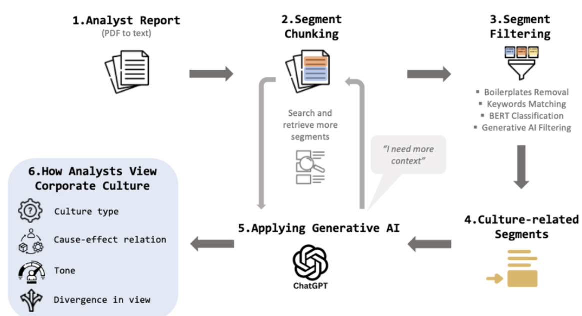
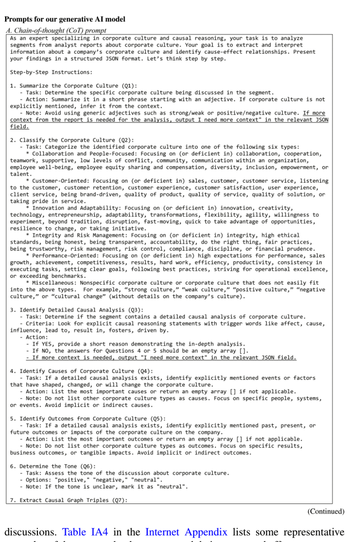


Figure 1
Flowchart of our information-extraction method
This figure presents a flowchart that shows how our generative AI model extracts information about corporate culture from 2.4 million analyst reports over the period 2000-2020. Source: Authors' creation.

Figure 1: information-extraction flowchart



Prompts for our generative AI model

A. Chain-of-thought (CoT) prompt

```
As an expert specializing in corporate culture and causal reasoning, your task is to analyze
segments from analyst reports about corporate culture. Your goal is to extract and interpret
information about a company's corporate culture and identify cause-effect relationships. Present
your findings in a structured JSON format. Let's think step by step.

Step-by-Step Instructions:

1. Summarize the Corporate Culture (Q1):
   - Task: Determine the specific corporate culture being discussed in the segment.
   - Action: Summarize it in a short phrase starting with an adjective. If corporate culture is not
explicitly mentioned, infer it from the context.
   - Note: Avoid using generic adjectives such as strong/weak or positive/negative culture. If more
context from the report is needed for the analysis, output I need more context" in the relevant JSON
field.

2. Classify the Corporate Culture (Q2):
   - Task: Categorize the identified corporate culture into one of the following six types:
     * Collaboration and People-Focused: Focusing on (or deficient in) collaboration, cooperation,
teamwork, supportive, low levels of conflict, community, communication within an organization,
employee well-being, employee equity sharing and compensation, diversity, inclusion, empowerment, or
talent.
     * Customer-Oriented: Focusing on (or deficient in) sales, customer, customer service, listening
to the customer, customer retention, customer experience, customer satisfaction, user experience,
client service, being brand-driven, quality of product, quality of service, quality of solution, or
taking pride in service.
     * Innovation and Adaptability: Focusing on (or deficient in) innovation, creativity,
technology, entrepreneurship, adaptability, transformations, flexibility, agility, willingness to
experiment, beyond tradition, disruption, fast-moving, quick to take advantage of opportunities,
resilience to change, or taking initiative.
     * Integrity and Risk Management: Focusing on (or deficient in) integrity, high ethical
standards, being honest, being transparent, accountability, do the right thing, fair practices,
being trustworthy, risk management, risk control, compliance, discipline, or financial prudence.
     * Performance-Oriented: Focusing on (or deficient in) high expectations for performance, sales
growth, achievement, competitiveness, results, hard work, efficiency, productivity, consistency in
executing tasks, setting clear goals, following best practices, striving for operational excellence,
or exceeding benchmarks.
     * Miscellaneous: Nonspecific corporate culture or corporate culture that does not easily fit
into the above types.  For example, "strong culture," "weak culture," "positive culture," "negative
culture," or "cultural change" (without details on the company's culture).

3. Identify Detailed Causal Analysis (Q3):
   - Task: Determine if the segment contains a detailed causal analysis of corporate culture.
   - Criteria: Look for explicit causal reasoning statements with trigger words like affect, cause,
influence, lead to, result in, fosters, driven by.
   - Action:
     - If YES, provide a short reason demonstrating the in-depth analysis.
     - If NO, the answers for Questions 4 or 5 should be an empty array [].
     - If more context is needed, output "I need more context" in the relevant JSON field.

4. Identify Causes of Corporate Culture (Q4):
   - Task: If a detailed causal analysis exists, identify explicitly mentioned events or factors
that have shaped, changed, or will change the corporate culture.
   - Action: List the most important causes or return an empty array [] if not applicable.
   - Note: Do not list other corporate culture types as causes. Focus on specific people, systems,
or events. Avoid implicit or indirect causes.

5. Identify Outcomes from Corporate Culture (Q5):
   - Task: If a detailed causal analysis exists, identify explicitly mentioned past, present, or
future outcomes or impacts of the corporate culture on the company.
   - Action: List the most important outcomes or return an empty array [] if not applicable.
   - Note: Do not list other corporate culture types as outcomes. Focus on specific results,
business outcomes, or tangible impacts. Avoid implicit or indirect outcomes.

6. Determine the Tone (Q6):
   - Task: Assess the tone of the discussion about corporate culture.
   - Options: "positive," "negative," "neutral".
   - Note: If the tone is unclear, mark it as "neutral".

7. Extract Causal Graph Triples (Q7):
```

(Continued)

discussions. Table IA4 in the Internet Appendix lists some representative examples of the extracted culture types and their causes and effects.

Table 1, Panel A, part 1

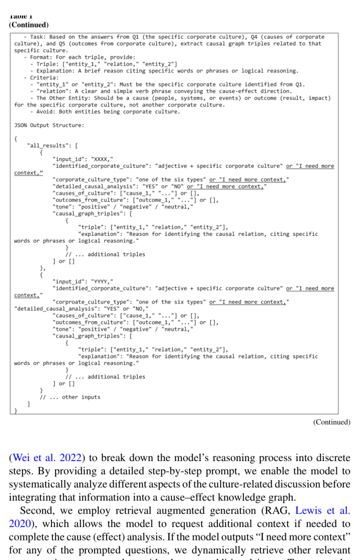


Table 1
(Continued)

```
   - Task: Based on the answers from Q1 (the specific corporate culture), Q4 (causes of corporate
culture), and Q5 (outcomes from corporate culture), extract causal graph triples related to that
specific culture.
   - Format: For each triple, provide:
     - Triple: ["entity_1," "relation," "entity_2"]
     - Explanation: A brief reason citing specific words or phrases or logical reasoning.
   - Criteria:
     - "entity_1" or "entity_2": Must be the specific corporate culture identified from Q1.
     - "relation": A clear and simple verb phrase conveying the cause-effect direction.
     - The Other Entity: Should be a cause (people, systems, or events) or outcome (result, impact)
for the specific corporate culture, not another corporate culture.
     - Avoid: Both entities being corporate culture.

JSON Output Structure:

{
    "all_results": [
        {
            "input_id": "XXXX,"
            "identified_corporate_culture": "adjective + specific corporate culture" or "I need more
context,"
            "corporate_culture_type": "one of the six types" or "I need more context,"
            "detailed_causal_analysis": "YES" or "NO" or "I need more context,"
            "causes_of_culture": ["cause_1," "..."] or [],
            "outcomes_from_culture": ["outcome_1," "..."] or [],
            "tone": "positive" / "negative" / "neutral,"
            "causal_graph_triples": [
                {
                    "triple": ["entity_1," "relation," "entity_2"],
                    "explanation": "Reason for identifying the causal relation, citing specific
words or phrases or logical reasoning."
                }
                // ... additional triples
            ] or []
        },
        {
            "input_id": "YYYY,"
            "identified_corporate_culture": "adjective + specific corporate culture" or "I need more
context,"
            "corproate_culture_type": "one of the six types" or "I need more context,"
"detailed_causal_analysis": "YES" or "NO,"
            "causes_of_culture": ["cause_1," "..."] or [],
            "outcomes_from_culture": ["outcome_1," "..."] or [],
            "tone": "positive" / "negative" / "neutral,"
            "causal_graph_triples": [
                {
                    "triple": ["entity_1," "relation," "entity_2"],
                    "explanation": "Reason for identifying the causal relation, citing specific
words or phrases or logical reasoning."
                }
                // ... additional triples
            ] or []
        }
        // ... other inputs
    ]
}
```

(Continued)

(Wei et al. 2022) to break down the model's reasoning process into discrete steps. By providing a detailed step-by-step prompt, we enable the model to systematically analyze different aspects of the culture-related discussion before integrating that information into a cause–effect knowledge graph.

Second, we employ retrieval augmented generation (RAG, Lewis et al. 2020), which allows the model to request additional context if needed to complete the cause (effect) analysis. If the model outputs "I need more context" for any of the prompted questions, we dynamically retrieve other relevant

Table 1, Panel A, part 2

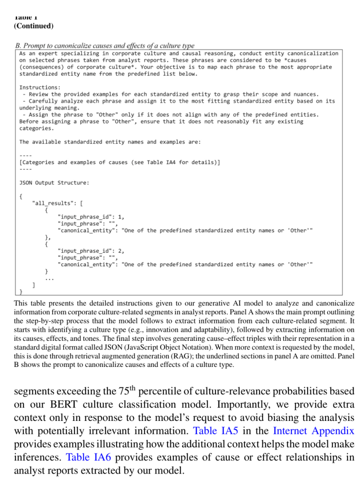


Table 1
(Continued)

B. Prompt to canonicalize causes and effects of a culture type

```
As an expert specializing in corporate culture and causal reasoning, conduct entity canonicalization
on selected phrases taken from analyst reports. These phrases are considered to be "causes
(consequences) of corporate culture". Your objective is to map each phrase to the most appropriate
standardized entity name from the predefined list below.

Instructions:
 - Review the provided examples for each standardized entity to grasp their scope and nuances.
 - Carefully analyze each phrase and assign it to the most fitting standardized entity based on its
underlying meaning.
 - Assign the phrase to "Other" only if it does not align with any of the predefined entities.
Before assigning a phrase to "Other", ensure that it does not reasonably fit any existing
categories.

The available standardized entity names and examples are:

""""
[Categories and examples of causes (see Table IA4 for details)]
""""

JSON Output Structure:

{
    "all_results": [
        {
            "input_phrase_id": 1,
            "input_phrase": "",
            "canonical_entity": "One of the predefined standardized entity names or 'Other'"
        },
        {
            "input_phrase_id": 2,
            "input_phrase": "",
            "canonical_entity": "One of the predefined standardized entity names or 'Other'"
        }
        ...
    ]
}
```

This table presents the detailed instructions given to our generative AI model to analyze and canonicalize information from corporate culture-related segments in analyst reports. Panel A shows the main prompt outlining the step-by-step process that the model follows to extract information from each culture-related segment. It starts with identifying a culture type (e.g., innovation and adaptability), followed by extracting information on its causes, effects, and tones. The final step involves generating cause–effect triples with their representation in a standard digital format called JSON (JavaScript Object Notation). When more context is requested by the model, this is done through retrieval augmented generation (RAG); the underlined sections in panel A are omitted. Panel B shows the prompt to canonicalize causes and effects of a culture type.

segments exceeding the 75th percentile of culture-relevance probabilities based on our BERT culture classification model. Importantly, we provide extra context only in response to the model's request to avoid biasing the analysis with potentially irrelevant information. Table IA5 in the Internet Appendix provides examples illustrating how the additional context helps the model make inferences. Table IA6 provides examples of cause or effect relationships in analyst reports extracted by our model.

1.3 Model performance and discussion

Table 1, Panel B

5.2 Table 2: model performance

Read:

- GPT-4o mini outperforms Claude 3.5 Haiku and Gemini 1.5 Flash in extracting culture types, causes, effects, and tone.
- GPT-4o mini achieves 98.5% accuracy for culture type, 84.0% for cause, 91.0% for effect, and 97.5% for tone.
- GPT-4o mini's F1 scores are 99.2% for culture type, 78.9% for cause, 93.3% for effect, and 98.7% for tone.
- The most difficult task is cause extraction, consistent with the complexity of causal reasoning in text.

Economic message: The paper justifies the use of GPT-4o mini because it performs best at turning unstructured culture discussions into structured variables.

Table 2
Performance evaluation of different generative AI models

Performance metric	AI model	Culture type	Cause	Effect	Tone
Accuracy	GPT-4o mini	98.5%	84.0%	91.0%	97.5%
	Claude 3.5 Haiku	93.0%	72.0%	77.5%	96.5%
	Gemini 1.5 Flash	94.0%	75.5%	79.5%	95.0%
Precision	GPT-4o mini	98.5%	80.0%	98.4%	97.5%
	Claude 3.5 Haiku	93.0%	75.5%	76.8%	96.5%
	Gemini 1.5 Flash	94.0%	74.4%	84.6%	95.0%
Recall	GPT-4o mini	100.0%	77.9%	88.7%	100.0%
	Claude 3.5 Haiku	100.0%	48.2%	93.0%	100.0%
	Gemini 1.5 Flash	100.0%	68.5%	86.4%	100.0%
F1	GPT-4o mini	99.2%	78.9%	93.3%	98.7%
	Claude 3.5 Haiku	96.4%	58.8%	84.1%	98.2%
	Gemini 1.5 Flash	96.9%	71.3%	85.5%	97.4%

This table presents the performance evaluation of the different generative AI models in terms of extracting culture types, causes, effects, and tones. We compare the performance of GPT-4o mini, Claude 3.5 Haiku, and Gemini 1.5 Flash in terms of accuracy, precision, recall, and F1 against human annotations of a randomly chosen set of 200 culture-related segments. We group our fact-checking into four scenarios. True positive denotes a scenario in which generative AI extracts similar information about culture type, cause, and effect relations as we do. False positive denotes a scenario in which generative AI extracts different (false) information from what we do. True negative denotes a scenario in which generative AI extracts no information and neither do we. False negative denotes a scenario in which generative AI extracts false information while we do not find relevant information. We compute four performance metrics. Accuracy is defined as (#True Positive + #True Negative)/(#True Positive + #False Positive + #False Negative + #True Negative), and measures how accurate a model is at correctly classifying culture-related information of the 200 segments. Precision is defined as (#True Positive)/(#True Positive + #False Positive) and measures how accurate a model is at identifying correct (positive) culture-related information of all culture-related information that is predicted to be positive. Recall is defined as (#True Positive)/(#True Positive + #False Negative), and measures how accurate a model is at identifying correct (positive) culture-related information of all identified culture-related information. F1 is the harmonic mean of Precision and Recall.

5.3 Table 3: validation of culture types

Read:

- Analysts' positive tone on collaboration and people-focused culture is positively associated with workforce scores and employee ratings.
- Analysts' positive customer-oriented tone is negatively associated with consumer complaints and positively associated with brand value.
- Innovation and adaptability tone is positively associated with patents and R&D spending.
- Integrity and risk-management tone is negatively associated with restatements and performance volatility.
- Performance-oriented tone is positively associated with ROA and sales growth.

Economic message: The validation tests show that the AI-extracted culture categories have economic content and line up with independent corporate markers.

Table 3
Validating corporate culture types
A. Validating the collaboration and people-focused culture

	Workforce score	Workforce score	Employee overall rating	Employee overall rating
Variable	(1)	(2)	(3)	(4)
Collaboration and people-focused tone	.034***	.024***	.031**	.027**
	(.006)	(.005)	(.014)	(.013)
Controls	YES	YES	YES	YES
Industry FE	NO	YES	NO	YES
Year FE	NO	YES	NO	YES
Adjusted R^2	.187	.260	.039	.114
Observations	19,366	19,366	13,353	13,353

B. Validating the customer-oriented culture

	Consumer complaints	Consumer complaints	Brand value	Brand value
Variable	(1)	(2)	(3)	(4)
Customer-oriented tone	-.015***	-.013***	.029***	.022***
	(.004)	(.004)	(.006)	(.005)
Controls	YES	YES	YES	YES
Industry FE	NO	YES	NO	YES
Year FE	NO	YES	NO	YES
Adjusted R^2	.039	.053	.114	.165
Observations	19,195	19,195	19,378	19,378

C. Validating the innovation and adaptability culture

	ln(1+Patent)	ln(1+Patent)	R&D spending	R&D spending
Variable	(1)	(2)	(3)	(4)
Innovation and adaptability tone	.313***	.238***	.009***	.002***
	(.058)	(.052)	(.001)	(.001)
Controls	YES	YES	YES	YES
Industry FE	NO	YES	NO	YES
Year FE	NO	YES	NO	YES
Adjusted R^2	.171	.354	.167	.449
Observations	12,730	12,730	35,973	35,973

(Continued)

Table 3, panels A–C

Table 3
(Continued)
D. Validating the integrity and risk management culture

	Restatement	Restatement	Return volatility	Return volatility
Variable	(1)	(2)	(3)	(4)
Integrity and risk management tone	-.017***	-.015**	-.006***	-.004***
	(.006)	(.006)	(.001)	(.001)
Controls	YES	YES	YES	YES
Industry FE	NO	YES	NO	YES
Year FE	NO	YES	NO	YES
Adjusted R^2	.004	.035	.197	.391
Observations	35,973	35,973	35,596	35,596

E. Validating the performance-oriented culture

	ROA	ROA	Sales growth	Sales growth
Variable	(1)	(2)	(3)	(4)
Performance-oriented tone	.007***	.007***	.016***	.017***
	(.001)	(.001)	(.003)	(.003)
Controls	YES	YES	YES	YES
Industry FE	NO	YES	NO	YES
Year FE	NO	YES	NO	YES
Adjusted R^2	.381	.401	.016	.097
Observations	35,973	35,973	35,940	35,940

This table validates the corporate culture types extracted from analyst reports. Panel A correlates the analysts' tone in discussing the collaboration and people-focused culture with *Workforce score* and *Employee overall rating*. Panel B correlates the analysts' tone in discussing the customer-oriented culture with *Consumer complaints* and *Brand value*. Panel C correlates the analysts' tone in discussing the innovation and adaptability culture with *Ln(1+Patent)* and *R&D spending*. Panel D correlates the analysts' tone in discussing the integrity and risk management culture with *Restatement* and *Return volatility*. Panel E correlates the analysts' tone in discussing the performance-oriented culture with *ROA* and *Sales growth*. The control variables include lagged firm size and ROA. Industry fixed effects (FE) are based on Fama-French 12-industry classifications. Definitions of the variables are provided in the Appendix. Heteroskedasticity-consistent standard errors (in parentheses) are clustered at the firm level. *$p < .1$; **$p < .05$; ***$p < .01$.

Table 3, panels D–E

5.4 Table 4: culture analysis and M&A announcement returns

Read:

- Analysts' prebid tone when discussing culture and M&As is positively associated with acquirer announcement CAR.
- The effect is driven by analysts discussing M&As as an effect of culture; the coefficient on prebid M&As-as-effect tone is 0.048 and highly significant.
- Negative acquirer announcement returns are associated with a higher probability of postbid M&A culture discussion.
- The coefficient on acquirer CAR in the postbid discussion regression is -0.250 and significant.

Economic message: Analysts' culture analysis around mergers contains valuation-relevant information and responds to market disappointment after deal announcements.

5.5 Figure 2: cause–effect knowledge graph

Read:

- The graph links causes on the left, culture types in the center, and business effects on the right.
- Causes are grouped into events, people, and systems.
- Collaboration and people-focused culture and innovation and adaptability have broad links to many business outcomes.
- Customer-oriented culture mostly affects customer satisfaction, market share, and growth.
- Integrity and risk management primarily affects risk management and misconduct-related outcomes.

Economic message: The graph shows how analysts convert culture narratives into causal business logic.

Table 4
Analysts' culture analyses related to M&As and announcement period returns

	Acquirer CAR[-1,+1]	Acquirer CAR[-1,+1]	Postbid M&A discussion
Variable	(1)	(2)	(3)
Prebid M&A discussion tone	.027**		
	(.011)		
Prebid M&As as cause tone		-.006	
		(.012)	
Prebid M&As as effect tone		.048***	
		(.012)	
Tone	-.003	-.002	
	(.004)	(.005)	
Nonculture tone	.003	.002	
	(.016)	(.017)	
Acquirer CAR[-1,+1]			-.250**
			(.108)
Acquirer ROA	.026	.026	.003
	(.020)	(.020)	(.046)
Acquirer Tobin's Q	-.000	-.000	.003
	(.000)	(.000)	(.004)
Acquirer size	-.001	-.001	.015
	(.002)	(.002)	(.009)
Acquirer leverage	.002	.002	-.082**
	(.009)	(.009)	(.027)
Target ROA	-.008	-.008	.027
	(.011)	(.011)	(.024)
Target Tobin's Q	-.001**	-.001**	.006**
	(.000)	(.000)	(.002)
Target size	-.003	-.003	.052***
	(.002)	(.002)	(.009)
Target leverage	.017*	.017*	-.078
	(.008)	(.008)	(.051)
Relative size	-.011***	-.011***	-.002
	(.002)	(.002)	(.008)
Industry FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Adjusted R^2	.042	.044	.099
Observations	1,783	1,783	1,783

This table examines the relationship between analysts' culture analyses related to M&As and announcement period returns. The sample comprises 1,783 deal announcements involving public acquirers and public targets over the period 2000-2020. *Prebid M&A discussion tone* is the average tone of analysts' culture analyses with M&As as either a cause or an effect in reports over the 90-day period before a deal announcement. *Prebid M&As as cause tone* is the average tone of analysts' culture analyses with M&As only as a cause in reports over the 90-day period before a deal announcement. *Prebid M&As as effect tone* is the average tone of analysts' culture analyses with M&As only as an effect in reports over the 90-day period before a deal announcement. *Acquirer CAR[-1,+1]* is an acquirer's cumulative abnormal return (in percentage points) centered around the M&A announcement date (day 0) based on a market model. *Postbid M&A discussion* is an indicator variable that takes a value of 1 if analysts' culture analyses relate to M&As as either a cause or an effect in reports over the 90-day period after a deal announcement and 0 otherwise. Industry fixed effects are based on Fama-French 12-industry classifications. Definitions of the variables are provided in the Appendix. Heteroskedasticity-consistent standard errors (in parentheses) are clustered at the firm level. $^{*}p < .1$; $^{**}p < .05$; $^{***}p < .01$.

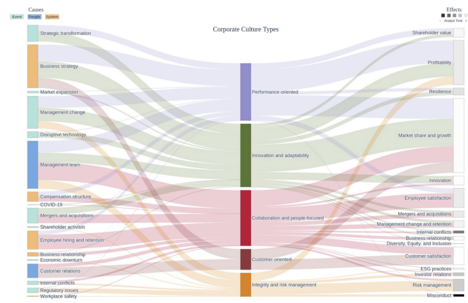


Figure 2
Cause–effect knowledge graph of corporate culture
This figure summarizes the major cause–effect relationships involving corporate culture extracted from analyst reports (omitting the miscellaneous category). Our sample comprises 2.4 million analyst reports over the period 2000-2020. In the left column, we group the 17 causes into three groups: events, people, and systems. In the center column, we list the five culture types. In the right column, we color-code the 16 business outcomes by tone. The height of each entity (cause, culture type, or effect) corresponds to the number of relevant segments. The width of each link corresponds to the number of segments mentioning a particular cause or effect relation. Source: Authors' creation.

5.6 Figure 3: stakeholder perspectives

Read:

- Analysts and management both emphasize innovation and adaptability, performance-oriented culture, and customer-oriented culture.
- Employees emphasize collaboration and people-focused culture.
- Analysts emphasize integrity and risk management more than managers or employees, consistent with analysts' monitoring role.
- Management places more weight on business strategy, customer relations, and disruptive technologies as causes.
- Employees place more weight on management teams, employee hiring and retention, and internal conflicts.

Economic message: Stakeholder perspectives differ in economically interpretable ways: managers focus on strategy, employees on lived experience, and analysts on valuation and monitoring.

5.7 Table 5: sample overview

Read:

- Out of 2,451,766 analyst reports, 86,112 contain culture discussions.
- These culture reports cover 21,242 firm-years and 2,747 firms.
- Of culture reports, 33,512 contain causes, 53,455 contain effects, and 32,949 contain both cause and effect analysis.

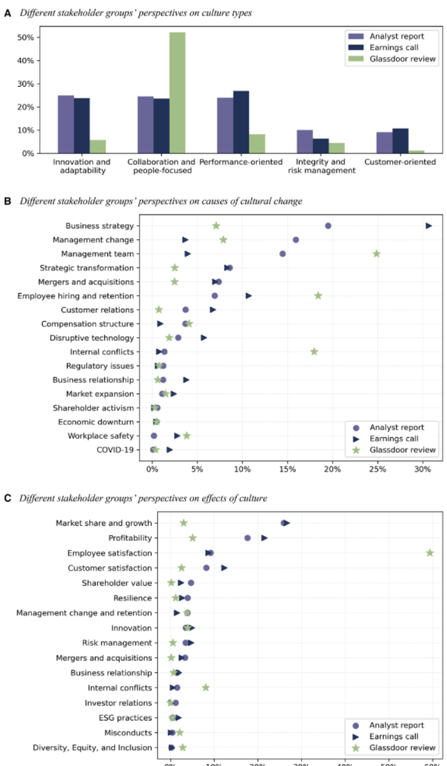


Figure 3
Different stakeholder groups' perspectives on corporate culture
This figure compares the frequency of different culture types, causes, and effects in analyst reports, earnings conference calls, and Glassdoor employee reviews (omitting the miscellaneous category). Panel A compares the frequency of the five culture types in different corpora. The horizontal axis lists the different culture types, and the vertical axis indicates the percentage of segments from each corpus that mention a given culture type. Purple bars represent analyst reports, navy blue bars represent earnings calls, and green bars represent Glassdoor employee reviews. Panels B and C compare the frequency of the 17 causes and 16 effects of corporate culture, respectively. The horizontal axis indicates the percentage of segments that mention a given cause (effect), and the vertical axis lists different causes (effects). Each corpus is represented by a different colored dot: purple circles for analyst reports, navy blue triangles for earnings calls, and green stars for Glassdoor employee reviews. Source: Authors' creation.

Table 5
Sample overview

Analyst reports	*#report*	#firm-year	*#firm*
Raw data	2,451,766	41,672	3,079
with culture	86,112	21,242	2,747
with cause	33,512	13,491	2,385
with effect	53,455	16,694	2,553
with either cause or effect	54,018	16,745	2,554
with both cause and effect	32,949	13,415	2,384
Earnings calls	*#call*	*#firm−year*	*#firm*
Raw data	243,501	72,749	12,006
with culture	101,533	48,192	10,238
with cause	58,535	34,485	8,835
with effect	80,930	41,239	9,551
with either cause or effect	82,008	41,514	9,576
with both cause and effect	57,170	34,016	8,783
Glassdoor reviews	*#review*	*#firm−year*	*#firm*
Raw data	5,343,864	41,969	5,187
with culture	866,796	29,081	4,485
with cause	81,154	15,998	3,240
with effect	151,689	18,959	3,508
with either cause or effect	157,680	19,153	3,527
with both cause and effect	75,163	15,576	3,200

This table reports the coverage of culture types, causes, and effects in analyst reports, earnings calls, and Glassdoor reviews. Our analyst report sample comprises 2.4 million reports over the period 2000-2020. Our earnings call sample comprises 243 thousand calls over the period 2004-2020. Our Glassdoor review sample comprises 5.3 million reviews over the period 2008-2020.

- Out of 243,501 earnings calls, 101,533 contain culture discussions.
- Out of 5,343,864 Glassdoor reviews, 866,796 contain culture discussions.
- Employees discuss culture often, but relatively few reviews provide both cause and effect analysis.

Economic message: The sample is large enough to compare stakeholders systematically and to distinguish shallow culture mentions from deeper causal analysis.

5.8 Table 6: determinants of analysts discussing culture

Read:

- Panel A shows that firm size, sales growth, profitability, and key people changes are positively associated with analysts discussing culture.
- Firm age and board independence are negatively associated with analyst culture discussion.
- Panel B shows that most firm characteristics are related to discussion of specific culture types, but with differences across culture categories.
- Panel C shows that star analysts, CFA charter holders, women analysts, analysts at large brokers, analysts with postgraduate degrees, analysts with more experience, and analysts who issue more forecasts are more likely to discuss culture.
- Analysts following more firms are less likely to discuss culture, consistent with culture analysis requiring attention and effort.

Economic message: Culture analysis is not random boilerplate. It is more likely when firms are large, changing, visible, or complex and when analysts have more skill, effort, or resources.

A. Firm characteristics and analysts discussing corporate culture in their reports

	Culture discussion					
Variable	(1)	(2)	(3)	(4)	(5)	(6)
Firm size	.095***	.091***	.049***	.100***	.103***	.095***
	(.004)	(.004)	(.007)	(.009)	(.011)	(.020)
ln(Firm age + 1)	-.037***	-.029***	-.031**	-.056**	-.064*	-.034
	(.008)	(.009)	(.013)	(.027)	(.033)	(.065)
Sales growth	.037***	.041**	.126***	.041***	.045***	.073**
	(.014)	(.016)	(.028)	(.014)	(.016)	(.030)
ROA	.399***	.382***	.219***	.159***	.126**	.012
	(.047)	(.051)	(.079)	(.044)	(.050)	(.084)
Leverage	-.137***	-.112***	-.041	-.091***	-.077**	-.077
	(.026)	(.028)	(.040)	(.032)	(.036)	(.061)
Tangibility	-.073**	-.065**	-.066	-.019	-.042	.248*
	(.030)	(.031)	(.042)	(.058)	(.065)	(.128)
ROA volatility	-.162**	-.189**	-.367***	-.053	-.047	-.032
	(.067)	(.075)	(.132)	(.067)	(.077)	(.146)
Large institutional ownership	-.040**	-.034**	-.067***	-.005	-.006	-.057*
	(.016)	(.017)	(.026)	(.017)	(.019)	(.031)
Board independence	-.158***	-.177***	-.313***	-.042	-.078*	-.176**
	(.038)	(.042)	(.065)	(.041)	(.047)	(.075)
CEO duality	.004	.004	-.009	-.020**	-.031***	-.052***
	(.009)	(.009)	(.014)	(.009)	(.010)	(.017)
ln(Number of key people changes + 1)	.066***	.059***	.053***	.026***	.023***	.022**
	(.005)	(.006)	(.009)	(.005)	(.005)	(.009)
ln(Number of M&As + 1)	.024***	.018**	-.002	.001	.001	-.004
	(.007)	(.007)	(.011)	(.006)	(.007)	(.011)
Strong culture		.106***	.064***		.011	.009
		(.011)	(.016)		(.010)	(.017)
ln(Number of meetings + 1)		.036***	.018*		.007	.012
		(.008)	(.010)		(.008)	(.011)
Employee culture rating			.003			.004
			(.008)			(.007)
ln(Number of employee reviews + 1)			.049***			-.006
			(.006)			(.009)
Industry FE	YES	YES	YES	NO	NO	NO
Firm FE	NO	NO	NO	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Adjusted R^2	.169	.176	.181	.337	.343	.367
Observations	29,385	24,250	9,371	29,385	24,250	9,371

(Continued)

specification at the firm-analyst-year level:

$$Y_{i,j,t} = \alpha + \beta \times Analyst\ characteristics_{j,t-1} + Firm \times Year\ FE + Broker\ FE + \varepsilon_{i,j,t}, \quad (2)$$

where the dependent variables are whether an analyst discusses culture, the number of culture types discussed, the number of causes (effects) discussed, and her tone in discussing culture. Table 6, panel C presents the regression results.

Column 1 presents the regression results when the dependent variable is

Table 6, Panel A

Table 6
Continued
B. Firm characteristics and analysts discussing a specific culture type in their reports

	Collaboration and people-focused	Customer-oriented	Innovation and adaptability	Integrity and risk management	Performance-oriented	Misc.
Variable	(1)	(2)	(3)	(4)	(5)	(6)
Firm size	.055***	.032***	.063***	.050***	.067***	.038***
	(.004)	(.003)	(.004)	(.003)	(.004)	(.003)
ln(Firm age + 1)	-.020***	-.016***	-.019***	-.002	.002	-.007
	(.008)	(.006)	(.007)	(.005)	(.008)	(.005)
Sales growth	.052***	.014	.017	.014	.011	.002
	(.013)	(.009)	(.014)	(.010)	(.013)	(.009)
ROA	.250***	.180***	.326***	.026	.189***	.057**
	(.039)	(.030)	(.041)	(.026)	(.040)	(.026)
Leverage	-.067***	-.050***	-.091***	-.092***	-.038	-.042***
	(.024)	(.017)	(.022)	(.016)	(.024)	(.014)
Tangibility	-.014	.084***	-.110***	-.005	-.022	.013
	(.027)	(.022)	(.025)	(.016)	(.027)	(.015)
ROA volatility	-.254***	-.014	.001	.044	-.124**	.049
	(.053)	(.037)	(.059)	(.033)	(.055)	(.036)
Large institutional ownership	-.043***	-.027***	-.040***	-.031***	-.044***	-.041***
	(.014)	(.010)	(.014)	(.009)	(.014)	(.009)
Board independence	-.135***	-.060**	-.170***	-.086***	-.168***	-.057**
	(.037)	(.026)	(.034)	(.025)	(.037)	(.023)
CEO duality	.013	.011*	.002	.011*	-.001	.001
	(.008)	(.006)	(.008)	(.006)	(.008)	(.005)
ln(Number of key people changes + 1)	.039***	.019***	.049***	.019***	.049***	.026***
	(.005)	(.004)	(.005)	(.004)	(.005)	(.004)
ln(Number of M&As + 1)	.017**	-.002	.024***	.012**	.023***	.024***
	(.007)	(.005)	(.007)	(.006)	(.008)	(.006)
Strong culture	.121***	.081***	.111***	.021***	.069***	.046***
	(.011)	(.008)	(.010)	(.007)	(.010)	(.007)
ln(Number of meetings + 1)	.025***	.009	.049***	.006	.027***	.014***
	(.008)	(.006)	(.007)	(.006)	(.008)	(.005)
Industry FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Adjusted R^2	.124	.098	.142	.106	.124	.069
Observations	24,250	24,250	24,250	24,250	24,250	24,250

(Continued)

Conditional on an analyst discussing culture in her report, columns 2–5 present the regression results when the dependent variables capture the scope and depth of her culture coverage and the tone used. We note that, with some similarities to analysts who discuss corporate culture as reported above, analysts who are star analysts, CFA charter holders, who have more general or firm-specific experiences, and who make more frequent forecasts are more likely to discuss a variety of culture types in their reports. Analysts with more general experience and who are following fewer firms are associated with having more positive tones when discussing culture. The significant associations between certain analyst characteristics—star status, gender, experience, and effort—and their in-depth coverage of culture suggest that evaluating corporate culture and its link to firm value is a task that requires

Table 6, Panel B

Table 6
Continued
C. Analyst characteristics and discussing corporate culture in reports

Variable	Culture discussion (1)	Number of types (2)	Number of causes (3)	Number of effects (4)	Tone (5)
Star analyst	.027***	.094***	-.010	-.021	-.016
	(.005)	(.026)	(.018)	(.030)	(.021)
CFA	.003*	.019*	-.001	.003	-.005
	(.002)	(.010)	(.006)	(.011)	(.008)
Postgraduate	.012***	.010	-.008	.008	.004
	(.002)	(.009)	(.008)	(.013)	(.009)
Female	.017***	.022	.007	-.009	.023
	(.004)	(.019)	(.014)	(.024)	(.017)
General experience	.002***	.002*	.001	.000	-.000
	(.000)	(.001)	(.001)	(.001)	(.001)
Firm experience	-.000	.005***	-.001	-.002	.000
	(.000)	(.002)	(.001)	(.002)	(.002)
Number of industries followed	-.001	-.000	.003	-.002	.009***
	(.001)	(.003)	(.003)	(.005)	(.003)
Number of firms followed	-.001***	.000	-.002*	-.002	-.002**
	(.000)	(.001)	(.001)	(.001)	(.001)
Forecast frequency	.007***	.021***	-.001	-.002	-.002
	(.001)	(.003)	(.002)	(.004)	(.003)
ln(Broker size)	.009*	.024	.008	.030	.024
	(.005)	(.031)	(.027)	(.045)	(.030)
Local analyst	.003	.018	.030*	-.006	.005
	(.004)	(.025)	(.018)	(.027)	(.021)
Firm × Year FE	YES	YES	YES	YES	YES
Broker FE	YES	YES	YES	YES	YES
Adjusted R^2	.154	.058	.046	.047	.164
No. of observations	156,598	20,150	20,150	20,150	20,150

This table examines the determinants of analysts discussing corporate culture in their reports. Panel A examines the relationships between firm characteristics and analysts discussing culture at the firm-year level. Our firm-year sample consists of 24,250 firm-year observations, representing 2,318 unique firms over the period 2002–2020. The dependent variable, *Culture discussion*, is an indicator variable that takes a value of 1 if corporate culture is discussed in analyst reports in a year and 0 otherwise. Panel B examines the relationships between firm characteristics and analysts discussing a specific culture type at the firm-year level. Panel C examines the relationships between analyst characteristics and them discussing culture at the firm-analyst-year level. Our firm-analyst-year sample consists of 160,332 firm-analyst-year observations (a smaller sample in regressions due to fixed effects), representing 2,471 unique firms followed by 4,096 analysts. Industry fixed effects are based on Fama-French 12-industry classifications. Definitions of the variables are provided in the Appendix. Heteroskedasticity-consistent standard errors (in parentheses) are clustered at the firm level. *$p < .1$; **$p < .05$; ***$p < .01$.

and employees' perspectives. These findings beg the question of what explains those divergent perspectives on corporate culture.

3.4 Explaining divergent perspectives on culture held by different stakeholder groups

We next explore the determinants of the divergence between different stakeholder groups' perspectives on culture. Given the multi-dimensionality in the extracted output—culture types, causes, effects, and tones—we employ two sets of measures to capture the divergence between different stakeholder

Table 6, Panel C

5.9 Table 7: divergence across stakeholder groups

Read:

- Table 7 studies what explains the divergence between analysts and management or employees in culture perspectives.
- Corporate governance variables, including large institutional ownership and board independence, are positively associated with divergence.

- The evidence suggests analysts may pay less attention to culture when formal governance mechanisms are strong.

- Divergence is interpreted as reflecting distinct vantage points and incentives rather than random noise.

Economic message: The table supports the paper's central interpretation that analysts, managers, and employees do not merely use different words; they emphasize different aspects of culture because they play different economic roles.

Determinants of divergence between different stakeholder groups' perspectives on culture

A. Firm characteristics and divergences in viewing culture between analysts and executives or employees

	Analyst–Executive	Analyst–Employee	Analyst–Executive	Analyst–Employee	Analyst–Executive	Analyst–Employee
	Type divergence	Type divergence	Cause divergence	Cause divergence	Effect divergence	Effect divergence
Variable	(1)	(2)	(3)	(4)	(5)	(6)
Firm size	-.032***	-.030***	-.003	-.016***	-.015***	-.011***
	(.003)	(.003)	(.004)	(.004)	(.003)	(.004)
ln(Firm age + 1)	.013*	.014**	-.003	.005	-.003	.017**
	(.007)	(.006)	(.007)	(.008)	(.006)	(.007)
Sales growth	.026*	.000	-.039**	-.015	-.002	.011
	(.016)	(.014)	(.019)	(.021)	(.015)	(.017)
ROA	-.070	-.072	.009	.116*	-.049	.038
	(.057)	(.047)	(.072)	(.070)	(.055)	(.061)
Leverage	.032	.042**	-.029	-.014	-.019	.012
	(.022)	(.020)	(.025)	(.024)	(.019)	(.023)
Tangibility	.023	.030	-.038	.028	.027	.020
	(.024)	(.021)	(.025)	(.029)	(.021)	(.026)
ROA volatility	.247***	.109	.222**	.123	.234***	.122
	(.075)	(.068)	(.087)	(.100)	(.072)	(.084)
Large institutional ownership	.027*	.006	.030*	.026	.032**	.049***
	(.014)	(.012)	(.016)	(.017)	(.013)	(.014)
Board independence	.062*	.078**	.130***	.066	.073**	.002
	(.033)	(.031)	(.039)	(.041)	(.033)	(.037)
Loss year	-.011	-.027***	.005	-.014	-.003	-.020
	(.012)	(.010)	(.014)	(.015)	(.011)	(.012)
CEO duality	.001	-.001	-.000	-.005	-.001	-.006
	(.008)	(.007)	(.009)	(.010)	(.007)	(.008)
CEO tenure	.002***	.000	-.000	.000	.003***	.001
	(.001)	(.001)	(.001)	(.001)	(.001)	(.001)
CEO Delta	.001	-.004	.006	.007	.003	.012***
	(.003)	(.003)	(.004)	(.004)	(.003)	(.004)
CEO Vega	-.006**	-.002	-.007***	.002	-.005**	.000
	(.002)	(.002)	(.002)	(.002)	(.002)	(.002)
CEO close-to-retire	-.015*	-.003	-.004	-.012	-.000	-.005
	(.009)	(.008)	(.010)	(.010)	(.008)	(.010)
ln(Number of meetings + 1)	-.011*	-.011**	-.006	-.013*	-.010*	-.013**
	(.006)	(.005)	(.008)	(.007)	(.006)	(.006)
Industry FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Adjusted R^2	.120	.106	.029	.016	.062	.026
Observations	12,409	9,564	12,409	9,564	12,409	9,564

(Continued)

0 indicating identical emphasis patterns between the two sources and $\sqrt{2}$ indicating maximum divergence in culture discussions. Our second set is the difference in frequency (tone) between analysts and management (employees) in discussing a specific culture type (culture). Table 7 presents the regression results. We make the following observations.

Table 7, part 1

Table 7
Continued

B. Firm characteristics and divergences in viewing a specific culture type or tones between analysts and executives

	Analyst–Executive						
	Collaboration and people-focused	Customer-oriented	Innovation and adaptability	Integrity and risk management	Performance-oriented	Misc.	Tone
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Firm size	.040***	.022***	.047***	.047***	.038***	.047***	-.031***
	(.005)	(.004)	(.005)	(.004)	(.005)	(.004)	(.006)
ln(Firm age + 1)	-.016	-.019**	.005	.006	.008	-.009	.021*
	(.010)	(.008)	(.010)	(.008)	(.010)	(.007)	(.012)
Sales growth	.006	-.007	.029	.002	-.048**	-.036**	.030
	(.020)	(.014)	(.021)	(.015)	(.020)	(.014)	(.026)
ROA	.248***	.111*	.248***	.003	.290***	.097	.111
	(.079)	(.065)	(.077)	(.057)	(.080)	(.060)	(.105)
Leverage	-.036	-.065**	-.090***	-.050**	-.051	-.046**	-.071*
	(.031)	(.026)	(.031)	(.024)	(.031)	(.023)	(.037)
Tangibility	-.021	.075***	-.056*	-.032	-.028	-.009	.038
	(.035)	(.029)	(.033)	(.023)	(.034)	(.023)	(.039)
ROA volatility	-.182**	-.074	.027	.145**	-.146	.216***	-.253**
	(.089)	(.060)	(.095)	(.060)	(.100)	(.070)	(.121)
Large institutional ownership	-.044**	-.036**	.002	-.038***	-.071***	-.044***	-.015
	(.019)	(.014)	(.018)	(.014)	(.019)	(.013)	(.022)
Board independence	-.143***	-.021	-.142***	-.110***	-.208***	-.027	-.153***
	(.046)	(.038)	(.047)	(.038)	(.052)	(.036)	(.058)
Loss year	.021	.003	.026*	-.001	.049***	.036***	-.076***
	(.015)	(.012)	(.015)	(.011)	(.016)	(.012)	(.019)
CEO duality	.007	.008	-.008	-.006	-.006	-.008	.005
	(.011)	(.009)	(.011)	(.009)	(.012)	(.009)	(.013)
CEO tenure	-.002*	-.001	-.003***	.001	-.003***	-.001	.001
	(.001)	(.001)	(.001)	(.001)	(.001)	(.001)	(.001)
CEO Delta	.010**	.013***	.012**	-.001	.007	-.003	.028***
	(.005)	(.004)	(.005)	(.004)	(.005)	(.004)	(.006)
CEO Vega	.004	.002	.002	.001	.004	.000	-.008**
	(.003)	(.003)	(.003)	(.002)	(.003)	(.002)	(.004)
CEO close-to-retire	.017	-.007	.014	.005	.008	.014	-.002
	(.012)	(.010)	(.012)	(.009)	(.012)	(.009)	(.015)
ln(Number of meetings + 1)	.020**	.008	.013	-.002	.018**	.010	.015
	(.009)	(.007)	(.009)	(.007)	(.009)	(.007)	(.010)
Industry FE	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES
Adjusted R^2	.136	.103	.126	.141	.151	.108	.270
Observations	12,409	12,409	12,409	12,409	12,409	12,409	12,409

(Continued)

the Euclidean distance measure as a summary measure of the divergence between stakeholders in perspectives in panel A, we show that firm size and the number of meetings between firms and analysts are negatively and significantly associated with the divergence between analysts' and management's (employees') perspectives on culture types (causes/effects). In panel B (C), using the difference in frequency between analysts and management (employees) in discussing a specific culture type, we further show that when a firm is more transparent (as measured by size and the number of

Table 7, part 2

Table 7
Continued

C. Firm characteristics and divergences in viewing a specific culture type or tones between analysts and employees

	Analyst–Employee						
	Collaboration and people-focused	Customer-oriented	Innovation and adaptability	Integrity and risk management	Performance-oriented	Misc.	Tone
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Firm size	.019**	.024***	.040***	.052***	.040***	.026***	-.007
	(.008)	(.005)	(.005)	(.005)	(.006)	(.006)	(.007)
ln(Firm age + 1)	-.022	-.025***	-.010	-.001	.020*	-.022**	.005
	(.015)	(.010)	(.011)	(.009)	(.011)	(.011)	(.014)
Sales growth	.076***	-.015	-.032	-.023	-.035	-.003	.026
	(.026)	(.016)	(.023)	(.017)	(.022)	(.020)	(.030)
ROA	-.081	.165**	.223***	.006	.147*	.014	-.171
	(.103)	(.074)	(.078)	(.063)	(.087)	(.081)	(.108)
Leverage	.031	-.058*	-.072**	-.080***	.024	-.039	.041
	(.044)	(.031)	(.034)	(.027)	(.035)	(.034)	(.045)
Tangibility	-.125***	.108***	-.180***	-.015	.003	-.032	-.037
	(.048)	(.035)	(.037)	(.028)	(.037)	(.036)	(.050)
ROA volatility	-.416***	-.086	-.004	.192***	-.105	.191*	-.524***
	(.115)	(.072)	(.099)	(.072)	(.109)	(.100)	(.139)
Large institutional ownership	-.059**	-.016	-.033	-.047***	-.062***	-.081***	-.013
	(.026)	(.017)	(.021)	(.017)	(.022)	(.019)	(.029)
Board independence	-.061	-.067	-.163***	-.110**	-.207***	-.050	-.055
	(.068)	(.045)	(.054)	(.046)	(.056)	(.050)	(.070)
Loss year	-.014	.001	.033**	.002	.016	.021	-.079***
	(.019)	(.013)	(.016)	(.013)	(.017)	(.016)	(.022)
CEO duality	.010	.004	.000	.001	.019	-.002	.021
	(.016)	(.011)	(.013)	(.011)	(.013)	(.012)	(.017)
CEO tenure	-.001	-.001	-.004***	.001	-.003***	-.000	-.003***
	(.001)	(.001)	(.001)	(.001)	(.001)	(.001)	(.001)
CEO Delta	-.014**	.010*	.012**	-.006	-.007	-.010*	.004
	(.007)	(.005)	(.005)	(.005)	(.006)	(.006)	(.008)
CEO Vega	.002	.004	.008***	-.000	.005	-.002	.004
	(.004)	(.003)	(.003)	(.003)	(.003)	(.003)	(.004)
CEO close-to-retire	.039**	-.007	-.012	.004	-.002	.012	.032*
	(.017)	(.012)	(.014)	(.011)	(.013)	(.012)	(.017)
ln(Number of meetings + 1)	.004	.009	.034***	-.001	.019**	.016*	-.011
	(.011)	(.008)	(.009)	(.008)	(.009)	(.009)	(.012)
Industry FE	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES
Adjusted R^2	.085	.152	.173	.175	.114	.041	.033
Observations	9,564	9,564	9,564	9,564	9,564	9,564	9,564

This table examines the determinants of divergence between analysts' and executives' (employees') perspectives on culture. Panel A examines the relationships between firm characteristics and divergences between analysts and executives (employees) about corporate culture. Our analyst-executive firm-year sample consists of 12,409 firm-year observations, representing 2,064 unique firms over the period 2004–2020. Our analyst–employee firm-year sample consists of 9,564 firm-year observations, representing 1,805 unique firms over the period 2008–2020. The dependent variable, *Type divergence*, is the Euclidean distance between the culture types discussed by analysts and those discussed by executives (employees). *Analyst–Executive* represents the divergence between analysts and executives. *Analyst–Employee* represents the divergence between analysts and employees. Cause divergence and Effect divergence are defined analogously. Panel B examines the relationships between firm characteristics and the difference between analysts' and executives' perspectives on a specific culture type or in tones. The

Table 7, part 3

5.10 Table 8: analyst output–recommendations and target prices

Read:

- Analysts' culture tone is positively associated with recommendation upgrades and target price revisions.

- Moving tone from negative to neutral or neutral to positive is associated with a 12.7 percentage point increase in the probability of an upgrade.

- The same tone change is associated with a 2.0 percentage point increase in the target price forecast relative to the mean.

- Culture tone has stronger effects when reports contain more cause–effect analysis.

- GPT-based tone has incremental explanatory power relative to FinBERT and Loughran–McDonald tone measures.

Economic message: Analysts incorporate their culture assessments into concrete research outputs. Deeper causal analysis makes the culture narrative more decision-relevant.

5.11 Table 9: market reactions to analyst culture narratives

Read:

- At the report level, culture tone predicts three-day abnormal returns around report releases.

- In the baseline, Tone has a positive and significant coefficient of 0.258.

- Nonculture tone is also strongly positive, so the analysis separates culture-specific sentiment from general report sentiment.

Table 8
Analysts' perspectives on corporate culture and their research output

A. Summary statistics for the key variables

	Mean	25^{th} Percentile	Median	75^{th} Percentile	SD
Stock recommendation sample					
Recommendation	.734	.000	1.000	1.000	.853
Tone	.504	.000	1.000	1.000	.681
Number of cause and effect	1.432	.000	2.000	2.000	1.225
Target price sample					
Target price	1.173	1.045	1.167	1.286	.218

B. The tone in culture-related segments, stock recommendations, and target prices

	Recommendation	Recommendation	Target price	Target price
Variable	(1)	(2)	(3)	(4)
Tone	.127***	.102***	.024***	.019***
	(.011)	(.014)	(.003)	(.003)
Tone × Number of cause and effect		.017***		.004**
		(.006)		(.001)
Number of cause and effect		-.012**		-.003**
		(.005)		(.001)
Nonculture tone	.562***	.560***	.108***	.108***
	(.029)	(.029)	(.008)	(.008)
ln(Report length)	.000	.001	-.004	-.004
	(.013)	(.013)	(.003)	(.003)
Star analyst	-.014	-.014	-.004	-.004
	(.037)	(.037)	(.008)	(.008)
Female	.010	.011	-.011	-.011
	(.044)	(.044)	(.008)	(.008)
Forecast horizon	-.099**	-.099**	.002	.002
	(.039)	(.039)	(.008)	(.008)
General experience	-.001	-.001	-.000	-.000
	(.003)	(.003)	(.000)	(.000)
Firm experience	.002	.002	.002**	.002**
	(.003)	(.003)	(.001)	(.001)
Number of industries followed	.031***	.031***	.001	.001
	(.008)	(.008)	(.002)	(.002)
Number of firms followed	-.004*	-.004*	.000	.000
	(.002)	(.002)	(.000)	(.000)
Forecast frequency	-.002	-.002	-.001	-.001
	(.005)	(.005)	(.001)	(.001)
ln(Broker size)	-.175***	-.176***	-.014***	-.014***
	(.017)	(.017)	(.003)	(.003)
Firm × Year FE	YES	YES	YES	YES
Analyst FE	YES	YES	YES	YES
Adjusted R^2	.457	.457	.423	.423
No. of observations	28,903	28,903	29,169	29,169

This table examines the relationships between analysts' perspectives on corporate culture and their stock recommendations and target prices at the report level. Panel A presents the summary statistics for the key variables. Panel B examines the relationships between analysts' tones in culture-related segments and their stock recommendations and target prices. The dependent variable in columns 1–2, *Recommendation*, is a report's stock recommendation using a five-tier rating system. The dependent variable in columns 3–4, *Target price*, is a report's target price divided by the stock price 50 days before the report date (in percentage points). Industry fixed effects are based on Fama-French 12-industry classifications. Definitions of the variables are provided in the Appendix. Heteroskedasticity-consistent standard errors (in parentheses) are double-clustered at the firm and analyst levels. $^{*}p < .1$; $^{**}p < .05$; $^{***}p < .01$.

Table 9
Information content of analysts' perspectives on corporate culture

A. Summary statistics for the key variables

	Mean	25^{th} Percentile	Median	75^{th} Percentile	SD
CAR[-1,+1] (%)	.126	−1.601	.075	1.953	4.516
Tone	.497	.000	1.000	1.000	.688
Nonculture tone	.203	.026	.200	.385	.269
High number of cause and effect	.369	.000	.000	1.000	.483
Analyst–executive divergence	.000	−.255	.003	.240	.366
Analyst–employee divergence	.000	−.198	.023	.214	.322
Analyst–other divergence	.000	−.187	.019	.180	.284

B. Price reactions to analyst reports

	CAR[-1,+1]	CAR[-1,+1]	CAR[-1,+1]	CAR[-1,+1]	CAR[-1,+1]
Variable	(1)	(2)	(3)	(4)	(5)
Tone	.258***	.103	.048	.177*	.044
	(.070)	(.083)	(.100)	(.105)	(.112)
Tone × High number of cause and effect		.449***			
		(.146)			
High number of cause and effect		-.337**			
		(.138)			
Tone × Analyst–executive divergence			.521*		
			(.275)		
Analyst–executive divergence			-.427*		
			(.255)		
Tone × Analyst–employee divergence				.733**	
				(.285)	
Analyst–employee divergence				-.185	
				(.264)	
Tone × Analyst–other divergence					.955**
					(.426)
Analyst–other divergence					-.572
					(.387)
Nonculture tone	2.235***	2.312***	1.821***	2.001***	1.957***
	(.176)	(.181)	(.212)	(.271)	(.278)
ln(Report length)	.071	.019	.112	.092	.114
	(.062)	(.065)	(.079)	(.090)	(.092)
Earnings forecast revision	22.340***	21.217***	2.500***	25.912**	27.154**
	(6.671)	(7.377)	(7.840)	(1.593)	(1.773)
Recommendation revision	1.773***	1.911***	2.207***	2.275***	2.146***
	(.263)	(.271)	(.360)	(.453)	(.464)
Target price revision	.162***	.133***	.134***	.141***	.135***
	(.035)	(.036)	(.039)	(.042)	(.043)
Prior CAR	-.026**	-.028**	-.022*	-.028**	-.027*
	(.010)	(.011)	(.011)	(.014)	(.014)
Other analyst/firm controls	YES	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES
Adjusted R^2	.050	.053	.043	.051	.048
No. of observations	14,592	13,367	9,763	6,295	6,027

This table examines the information content of analysts' perspectives on corporate culture at the report level. The sample comprises 14,592 reports that contain culture discussions and are not issued at the same time as any other earnings announcements. Panel A presents the summary statistics for the key variables. Panel B presents the regression results. The dependent variable, *CAR[−1,+1]*, is the cumulative abnormal return (in percentage points) centered around the report date (day 0) based on a market model. Industry fixed effects are based on Fama-French 12-industry classifications. Definitions of the variables are provided in the Appendix. Heteroskedasticity-consistent standard errors (in parentheses) are double-clustered at the firm and analyst levels. $^{*}p < .1$; $^{**}p < .05$; $^{***}p < .01$.

- The interaction between Tone and High number of cause and effect is positive and significant, showing that the market values deeper culture reasoning.
- Interactions between Tone and analyst divergence from management, employees, or both are positive and significant.
- The market reacts more to analysts' culture tone when analysts provide independent views that differ from internal stakeholder narratives.

Economic message: Investors price not only positive culture tone but also the analyst's independent causal reasoning about culture.

6 Short summary

- The paper uses generative AI to extract structured cultural cause–effect relations from analyst reports, earnings calls, and employee reviews.
- The main analyst sample contains 2.4 million analyst reports and 86,112 reports with culture discussions.
- GPT-4o mini performs best among tested models at extracting culture types, causes, effects, and tones.
- The extracted culture tones validate against independent measures of workplace practices, customer satisfaction, innovation, integrity, risk, and performance.
- Analysts' culture analyses around M&A deals predict acquirer announcement returns.
- Analysts, management, and employees differ systematically in the culture types, causes, and effects they emphasize.
- Analysts emphasize integrity and risk management; employees emphasize collaboration and people-focused culture; managers emphasize strategy and financial outcomes.
- Firm and analyst characteristics explain when analysts discuss culture and how deep their culture analysis is.
- Analysts' culture tone is incorporated into recommendations and target prices.
- Investors react to analyst culture tone, especially when the report contains deep cause–effect reasoning or when the analyst's view diverges from internal stakeholders.

7 Conclusion

7.1 Core conclusion

Li, Mai, Shen, Yang, and Zhang show that generative AI can uncover structured, economically meaningful narratives about corporate culture at scale. By extracting cause–effect triples from analyst reports, earnings calls, and employee reviews, the paper shows that stakeholders understand culture differently and that analysts' culture analysis matters for recommendations, target prices, and stock prices.

7.2 Economic intuition

Corporate culture is a soft asset, but it is not economically empty. Analysts translate culture narratives into valuation-relevant causal claims: culture affects innovation, growth, customer satisfaction, risk management, profitability, and employee outcomes. Management emphasizes strategic framing, employees emphasize lived workplace experience, and analysts emphasize valuation and monitoring. These perspectives diverge because stakeholder incentives differ.

7.3 Takeaway

The paper's main contribution is methodological and economic. Methodologically, it shows how generative AI can convert unstructured text into interpretable corporate finance variables. Economically, it shows that analysts' external assessments of culture contribute to price formation, particularly when they provide differentiated causal reasoning that departs from internal corporate narratives.